An exact counterexample to the Catalan–Larcombe–French ratio conjecture

Principia Math In Progress

11 September 2026

Abstract

We refute Conjecture 5.2 of Sun and Zhao. At the twelfth ratio iteration, the terms with indices $12, 13, 14$ violate log-convexity, although all three remain after the prescribed deletion of the first twelve terms. A finite integer-product certificate establishes the strict violation; no floating-point inequality is used in the proof.

Task 1: Resolve MathDB problem 332483

The Catalan–Larcombe–French numbers are given by $P_0 = 1$, $P_1 = 8$, and

$$n^2 P_n = 8(3n^2 - 3n + 1)P_{n-1} - 128(n-1)^2 P_{n-2} \qquad (n \geq 2). \tag{1}$$

For a positive sequence a, define $(\mathcal{R}a)_n = a_{n+1}/a_n$. The published conjecture [1, Conjecture 5.2, p. 601] asserts that, for each integer $k \geq 1$, the sequence $(\mathcal{R}^k P)_n$, after deleting its first k terms, is log-concave for odd k and log-convex for even k. Since its indexing begins at zero, the retained terms have $n \geq k$.

Theorem 1. *Put $b_n = (\mathcal{R}^{12}P)_n$. Then*

$$\frac{999999999814}{10^{12}} < \frac{b_{12}b_{14}}{b_{13}^2} < \frac{999999999815}{10^{12}} < 1. \tag{2}$$

Thus $b_{13}^2 > b_{12}b_{14}$, contrary to the required log-convexity of $(b_n)_{n\geq 12}$. The conjecture is false.

Converting the inequality to integer arithmetic

Let $(\Delta f)_n = f_{n+1} - f_n$. Induction gives

$$\log(\mathcal{R}^k a)_n = \Delta^k \log a_n.$$

Consequently

$$\log \frac{b_{12}b_{14}}{b_{13}^2} = \Delta^{14} \log P_{12} = \sum_{j=0}^{14} (-1)^j \binom{14}{j} \log P_{12+j}. \tag{3}$$

Define the positive integers

$$A = \prod_{\substack{0\leq j\leq 14 \\ j \text{ even}}} P_{12+j}^{\binom{14}{j}}, \qquad B = \prod_{\substack{0\leq j\leq 14 \\ j \text{ odd}}} P_{12+j}^{\binom{14}{j}}. \tag{4}$$

Equation (3) says exactly that $b_{12}b_{14}/b_{13}^2 = A/B$.

The finite certificate

The following table lists every input to (4). Each value follows by the recurrence (1); the signed exponent is $(-1)^{n-12}\binom{14}{n-12}$.

n	P_n	signed exponent
12	15008466534400	1
13	221460239482880	−14
14	3287994183188480	91
15	49074667327062016	−364
16	735814252604162048	1001
17	11076827875597352960	−2002
18	167338264534131408896	3003
19	2535936220929779040256	−3432
20	38539148242634697342976	3003
21	587172319637401677856768	−2002
22	8966541774166044673310720	1001
23	137211561613401637284478976	−364
24	2103704833567570984242774016	91
25	32310152363853898923615715328	−14
26	497042117469860910740195508224	1

Direct integer arithmetic gives the certificate

$$999999999814\,B < 10^{12}A < 999999999815\,B. \tag{5}$$

For complete reproducibility, the following Python program evaluates only integers, computes all the recurrence inputs, and verifies both comparisons. Python integers have arbitrary precision.

```
from math import comb, prod

P = [1, 8]
for n in range(2, 27):
    q = 8*(3*n*n-3*n+1)*P[n-1] - 128*(n-1)**2*P[n-2]
    assert q % (n*n) == 0
    P.append(q // (n*n))
A = prod(P[12+j]**comb(14,j) for j in range(0,15,2))
B = prod(P[12+j]**comb(14,j) for j in range(1,15,2))
assert 999999999814*B < 10**12*A < 999999999815*B
assert 999999999815 < 10**12
```

Both A and B have binary length 582535. The accompanying `verify_counterexample.py` retains hashes, the table, and certificate results in `certificate.json`; the proof depends on the exact integer comparisons, not on the hashes.

Proof of Theorem 1. All the table entries are positive, so all ratios used in the argument are defined. Divide (5) by the positive integer $10^{12}B$ and use (3)–(4). This gives (2). Since $k = 12$ is even and the offending triple has indices $12, 13, 14$, the source's required cutoff leaves the entire triple in the sequence. It therefore violates the exact published assertion. □

Verification and scope

The integer certificate is part of the proof. As additional checks, the verifier compares the recurrence values through $n = 26$ with the independent finite sum

$$P_n = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \binom{2j}{j}^2 2^{3n-4j},$$

and verifies all fifteen exponents by convolving the twelfth-difference coefficients with $(1, -2, 1)$. These checks pass exactly. The initial numerical search is retained separately as an experiment. No claim that $k = 12$ is the smallest possible failure is needed or made. The counterexample does not address a modified conjecture with a larger, order-dependent cutoff.

Task 2: Determine the correct eventual cutoff

For each $k \geq 1$, determine the least integer $N(k)$, if it exists, such that $(-1)^k \Delta^{k+2} \log P_n \geq 0$ for all $n \geq N(k)$. Prove effective bounds for $N(k)$ and obtain its growth as $k \to \infty$. The counterexample requires $N(12) \geq 13$.

Task 3: Classify failures at the proposed cutoff

Determine all k for which $(-1)^k \Delta^{k+2} \log P_k < 0$. Prove or refute that infinitely many such k exist, and develop exact certificates whose size remains practical as the iteration order grows.

Task 4: Explain the failure through the moment density

Use the arcsine-product representation $P_n/16^n = \int_0^1 u^n w(u)\, du$, where $w(u) = 4K(4u(1-u))/\pi^2$ and K uses its parameter convention. Derive uniform asymptotics for $\Delta^{k+2} \log P_n$ when n/k approaches a positive constant. Identify how the logarithmic singularity of w at $u = 1/2$ affects the sign near $n = k$.

Task 5: Classify cutoffs for symmetric beta products

Let X, Y be independent with densities proportional to $(1 - x^2)^{a-1}$ and $(1 - y^2)^{b-1}$ on $(-1, 1)$, where $a, b > 0$, and set $p_n = \mathbb{E}[((1 + XY)/2)^n]$. Determine which parameter pairs admit constants c, d such that $(-1)^k \Delta^{k+2} \log p_n \geq 0$ for every $k \geq 1$ and every integer $n \geq ck + d$. Find the sharp slope where it exists.

References

[1] B. Y. Sun and J. J. Y. Zhao, *Log-behavior of Two Sequences Related to the Elliptic Integrals*, Acta Mathematicae Applicatae Sinica, English Series **36** (2020), 590–602, Conjecture 5.2, p. 601. https://doi.org/10.1007/s10255-020-0952-1. Full publisher PDF: https://applmath.cjoe.ac.cn/jweb_yysxxb_en/EN/PDF/10.1007/s10255-020-0952-1. Current preprint: https://arxiv.org/abs/1602.04359.